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LLMs as learning-preference amplifiers in first-year higher education: a conceptual affordance framework and design implications

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Abstract

Purpose – This paper theorizes large language models (LLMs) as learning-preference amplifiers and introduces the AI Amplification Zone, specifying when LLM use strengthens productive learning strategies (e.g. strategic practice and reflection) versus entrenches maladaptive ones (e.g. avoidance and shallow processing).

Design/methodology/approach – A conceptual, theory-building synthesis integrating Affordance Theory with cognitive and motivational research. The analysis models a mechanism of preference detection → adaptation → ossification and derives testable propositions (P1–P3) and practice-oriented heuristics for first-year higher education.

Findings – The framework shows how desirable difficulties, clarified human–AI feedback roles, and UDL-aligned task structures can shift learners into the Amplification Zone, while guardrails (e.g. transparency, calibrated prompting and AI-free checkpoints) mitigate over-reliance. The paper advances three propositions: (P1) embedding desirable difficulties with LLM support promotes deeper learning strategies; (P2) scaffolded prompting plus human–AI co-teaching reduces over-reliance among learners with a strong performance orientation; and (P3) periodic AI-free checkpoints disrupt ossification loops and preserve transfer.

Research limitations/implications – As a conceptual synthesis, the framework requires empirical testing; the paper outlines falsifiable designs and evaluation measures (behavioral, performance and transfer) aligned to P1–P3.

Practical implications – Provides course-team heuristics: embed desirable difficulties, specify human/LLM feedback roles, schedule AI-free checkpoints and align tasks with UDL while monitoring over-reliance.

Social implications – Highlights equity and ethics considerations around profiling, transparency and bias; recommends guardrails to ensure AI augments rather than replaces productive struggle.

Originality/value – Reframes LLMs from generic assistants to preference amplifiers and offers a mechanism-level lens explaining why the same affordances can deepen learning for some students yet narrow strategy use for others.

Keywords Large language models, Generative AI, Affordances, Learning preferences, First-year students, Higher education, Instructional design, Human–AI co-teaching, Assessment design, AI literacy

Paper type Research article

Introduction

In the myth of Daedalus and Icarus, the gifted inventor constructs a pair of wings so that he and his son may escape captivity, not by brute force, but rather through harnessing the laws of nature. Yet, the tool he provides, so full of promise, becomes Icarus's undoing when misused.

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Soaring too high, too close to the sun, Icarus ignores his father's warnings and falls. Currently, education is at a pivotal crossroads that mirrors this myth. With generative AI tools now widely accessible, restricting students from using them is not only impractical but also misaligned with the realities of the world they are entering. This narrative resonates powerfully in the modern classroom, where new technologies, particularly generative artificial intelligence (AI), offer learners the freedom to shape their learning trajectories. For some students, these tools act as scaffolds: structuring their thoughts, clarifying ideas, or generating language that aligns with their cognitive preferences (Kim & Kim, 2022). For others, they may tempt shortcut thinking, masking underdeveloped skills behind the illusion of fluency (Wang *et al.*, 2023). Meanwhile, educators face the challenge of integrating AI in ways that support, rather than compromise, the development of essential skills. Table 1 provides an overview of illustrative examples of this.

Although AI use in education is expanding, its alignment with learning preferences remains underexplored. Emerging perspectives suggest that AI affects learners differently, amplifying existing cognitive preferences rather than serving all students equally. This amplification is not neutral; it can support or hinder learning depending on how affordances are used. For instance, some students use AI to organize ideas (Susilo, 2024), others for linguistic support like rephrasing or summarizing (Halkiopoulos & Gkintoni, 2024; Katiyar *et al.*, 2024), and visual learners for generating diagrams or comparisons (Kim & Kim, 2020). These patterns indicate that students often use AI in ways that reflect how they prefer to process information, even if they are not consciously aware of it. How students use AI can be better understood by looking at two ideas from education theory: cognitive preferences for learning, and affordances.

Table 1. Integration of AI in higher education

Integration area	Description and examples	Supporting evidence	Citations
Personalized Learning	AI adapts content, pace, and feedback to individual student needs and learning styles	Enhances engagement and learning outcomes through tailored experiences	Bhavani *et al.* (2025), Cai *et al.* (2025), Hooshyar *et al.* (2024), Jauhiainen and Garagorry Guerra (2024)
Intelligent Tutoring Systems	AI-powered tutors provide real-time guidance, answer questions, and scaffold learning	Supports student motivation, interaction, and academic performance	Bhavani *et al.* (2025), Létourneau *et al.* (2025), Mageira *et al.* (2022)
Automated Assessment and Feedback	AI grades assignments, quizzes, and provides instant, personalized feedback	Increases efficiency and provides timely, individualized support	Bhavani *et al.* (2025), Huang *et al.* (2025), Lee and Moore (2024)
Learning Analytics	AI analyzes student data to predict performance, identify at-risk students, and inform interventions	Enables data-driven decision-making and targeted support	Cai *et al.* (2025), Dann *et al.* (2024), Pan *et al.* (2024)
Interactive and Immersive Tools	AI powers chatbots, virtual reality, and collaborative platforms for engaging learning	Fosters communication, collaboration, and real-time feedback	Almusaed *et al.* (2023), Labadze *et al.* (2023), Wang and Xue (2024)
AI Literacy and Curriculum	AI is taught as a subject, preparing students to understand and use AI technologies	Prepares students for future workforce and technological advancements	Almatrafi *et al.* (2024), Bhavani *et al.* (2025)

Note(s): Illustrative, non-exhaustive overview of AI integration areas in higher education, with peer-reviewed examples (see *Citations*). Evidence claims summarize reported effects (e.g. timeliness of feedback, learning gains) rather than vendor assertions

Source(s): Authors' own work

Cognitive preferences are individual tendencies in processing information, including modes (verbal/visual), structure (sequential/global), and strategies (analytical/intuitive), influencing how learners approach tasks (Felder & Silverman, 1988; Riding & Rayner, 1998). Affordance Theory (AT) (e.g. Gibson, 1979; Norman, 1988) explains how learners perceive and engage with tools based on their goals and cognitive style. What a student finds useful depends on their goals and cognitive approach. When linked to learning preferences and AI, affordances emerge not as fixed properties, but as perceptions shaped by cognitive style, prior experience, and comfort with structure or ambiguity (see Conole & Dyke, 2004; Xue, Wang, & Muhaimaiti, 2023). Given this background, it is important to understand how students engage with AI and how educators can design learning experiences that accommodate this variation while still fostering critical thinking and avoiding over-reliance on shortcuts.

This conceptual, theory-building paper advances a focused framework for understanding how generative AI shapes first-year learning and translates that framework into concrete design implications for higher education. Specifically, it introduces the concept of AI as a learning-preference amplifier – not simply a tool that supports learning, but one that can *reinforce* learners' preferred approaches through repeated interaction. In this paper, *amplification* refers to the recursive process in which perceived "fit" between a learner's input and AI outputs strengthens preferred strategies over time, potentially increasing engagement but also narrowing flexibility. This differs from developmental scaffolding accounts such as the Zone of Proximal Development (ZPD; Vygotsky, 1978): rather than focusing on guided progression beyond current competence through targeted support, the amplification lens foregrounds how repeated ease and fluency can stabilize preferred strategies productive or maladaptive, through ongoing learner–tool interaction. Building on Affordance Theory, we examine how AI affordances invite particular actions and how those invitations interact with cognitive learning preferences (Felder & Silverman, 1988; Mayer & Massa, 2003). We further draw on sociomaterial perspectives that position AI as part of the learning environment rather than an external aid (Conole & Dyke, 2004; Fenwick, 2012). Using a conceptual model, visual framework, and summary matrix, we show how AI can support, bypass, or reshape engagement with cognitive demands in first-year learning, and we outline how educators can design learning experiences that encourage flexibility and reduce the risk of overdependence on AI tools.

Approach

This study follows a conceptual synthesis approach (Jaakkola, 2020; Whetten, 1989) to build an explanatory framework for how large language models (LLMs) can act as learning-preference amplifiers in first-year higher education. This is theory-building rather than an empirical study; the goal is to develop constructs and propose causal mechanisms leading to testable propositions (Lynham, 2002). Conceptual research draws on existing theoretical and empirical sources as its data (MacInnis, 2011).

Five strands were purposively explored: Affordance Theory (foundational and higher-education adaptations), cognitive and motivational research (e.g. desirable difficulties, metacognition, self-regulation), learning preferences and their critiques, first-year transition/retention, and LLM/AI in higher education (2023–2025). Selection prioritized works that illuminate mechanism-level links between tool affordances and learner behavior, alongside practice-oriented design scholarship (e.g. UDL). Conceptual rigor was maintained through *construct clarity* (Suddaby, 2010), *internal logical consistency* and *parsimony* (Whetten, 1989), and *explicit boundary specification* to first-year higher-education contexts. These measures ensure transparency and provide a coherent methodological foundation for the framework's theoretical and practical implications. To strengthen rigor, the corpus intentionally includes contrasting perspectives – for example, critiques of learning-style efficacy, concerns about automation bias and academic integrity, and equity-focused analyses – alongside studies reporting benefits.

The synthesis proceeded in four iterative stages: (1) *scoping and selection* of mechanism-relevant sources; (2) *analytic comparison* to identify recurring mechanisms of preference detection, adaptation, and ossification; (3) *integration* of these mechanisms through the vocabulary of Affordance Theory to develop the AI Amplification Zone framework; and (4) *derivation of propositions* (P1–P3) as falsifiable design claims for future empirical study. The scenarios used later in the manuscript are analytic vignettes designed to make the proposed mechanisms concrete across common first-year tasks (e.g. writing, coding, qualitative analysis). They are illustrative rather than evidential: they synthesize patterns documented in the cited literature and typical first-year learning designs, but they do not report new empirical data collected for this paper. We signpost these examples as “illustrative vignettes” to avoid conflating mechanism illustration with empirical claims.

As one may expect in a paper on this topic, a large language model–based assistant (e.g. ChatGPT) was used for language polishing and structured critical feedback on argument flow, sections and headings, and phrasing and titles. The model did not generate original scholarly content, data, analyses, or references, and all wording suggestions were reviewed, edited, and approved by the authors, who take full responsibility for the manuscript’s content. No confidential or non-public data were shared with the model. This disclosure aligns with the journal’s policy that LLMs are not authors and that any use should be documented in the manuscript.

Originality and contribution

This study offers a conceptual reframing of generative AI in higher education by theorizing LLMs as learning-preference amplifiers rather than generic assistants. The contribution is fourfold: (1) an integrative synthesis linking Affordance Theory with cognitive-motivational research to explain preference detection → adaptation → ossification; (2) an operational model – the AI Amplification Zone – that specifies conditions under which preference–affordance alignments promote deep learning or risk over-reliance; (3) a set of falsifiable design propositions (P1–P3) that translate the model into testable claims; and (4) practice-ready heuristics for first-year course and assessment design (e.g. calibrated desirable difficulties, human–AI co-teaching roles, and periodic AI-free checkpoints). Within the first-year context, this clarifies when and how personalization supports growth versus narrows strategy use, thereby extending affordance-based accounts with actionable design implications.

Learning preferences in first-year higher education

Cognitive preferences are mental tendencies that shape how learners approach tasks, such as favoring words over images, structure over openness, or step-by-step logic over holistic thinking (e.g. Felder & Silverman, 1988; Mayer & Massa, 2003). A critical stance in the literature emphasizes that *preference is not the same as effectiveness*: evidence does not support the claim that teaching exclusively to a preferred “learning style” improves learning outcomes. Instead, it is more useful to provide varied input and help learners become flexible in how they approach material (Cuevas, 2015; Kirschner & Van Merriënboer, 2013; Pashler, McDaniel, Rohrer, & Bjork, 2008). Consistent with prior syntheses, this study does not claim that teaching to a preferred style improves learning; rather, the analysis examines how LLM affordances interact with preferences to shape behavior.

For first-year university students, understanding learning preferences and techniques is particularly important, as the transition into higher education often requires adjusting to new academic expectations and unfamiliar learning environments. Preferences span familiar VARK modalities (Visual, Auditory, Read/Write, Kinesthetic) and broader tendencies like working solo or in groups, online or face-to-face. While shaped by background and context, most students benefit from multimodal approaches that blend styles, and moreover, learning preferences can become more flexible with time (Cameron & Rideout, 2020; Hernández-

Torrano, Ali, & Chan, 2017; Kennedy, Judd, Churchward, Gray, & Krause, 2008; Lujan & DiCarlo, 2006; Opdecam, Everaert, Keer, & Buysschaert, 2013; Ramteke & Meshram, 2017).

Learning preference diversity can be challenging for educators, as lectures often cater to a few styles, disengaging students who learn best through interaction or sensory experience. In highly diverse classrooms, demographic, academic and cultural differences can further complicate efforts to design inclusive learning criteria (Lim, 2024). As a result, many advocate for integrated and varied teaching methods, combining lectures, group work, practical activities, and different types of media to better support a wide range of learners (Al-Saud, 2013; Babyuk, Aksyonova, & Chebykina, 2019; Nasir, Mughal, & Rind, 2021; Smith, Carter-Rogers, Norris, & Brophy, 2022; Stöckert & Bogner, 2019). For first-year students in particular, such flexibility can help ease the transition to university-level study and increase their chances of academic success.

While understanding learning preferences is valuable, accommodating them in traditional classroom settings presents several practical challenges. These barriers can limit student engagement and may disadvantage those whose preferred ways of learning are not addressed. Key challenges include:

(1) Time and curriculum constraints that make varied teaching methods hard to implement.

(2) Rigid assessment formats that favor verbal and written expression.

(3) Limited access to flexible tools or physical spaces for group work or hands-on learning.

(4) Institutional pressure to standardize teaching and assessment.

(5) Students' lack of self-awareness or language to express their learning preferences.

These factors make it difficult to meet the diverse needs of first-year students through traditional methods and tools alone. As such, the emergence of AI tools introduces a new opportunity: allowing students to adapt learning processes themselves in ways that better match how they think and engage.

Cognitive affordances and AI

Affordance Theory describes how tools invite certain actions; in education, cognitive affordances refer to how tools support thinking, learning and problem-solving (Gibson, 1979; Bower, 2008; Norman, 1999). Affordances are relational, emerging from the interaction between person and tool, not purely objective or subjective. This aligns with sociomaterial perspectives that view learning as embedded in everyday practices involving both people and materials. This provides an interesting link to Practice Studies, which emphasize that social life is constituted through practices: routinized behaviors, sayings, and doings (Schatzki, 2001; Reckwitz, 2002). Practice Studies' sociomaterial perspective holds that tools such as materials and artifacts are entangled in practice (Fenwick, 2012). Rejecting strict subject-object dualism, this view views meaningful action as emerging through the interaction between actors and their material environment. In a comparable manner, affordances are also reinterpreted as relational and dynamic, extending beyond Gibson's (1979) original concept. Scholars such as Faraj and Azad (2012), Zammuto, Griffith, Majchrzak, Dougherty, and Faraj (2007), and Fayard and De Sanctis (2005) connect affordance theory directly to sociomaterial practice.

Generative AI offers a rich set of affordances because it responds dynamically to user input, allows for social and linguistic interaction, and may support conceptualizing and materializing concepts and products, allowing students to guide and shape its outputs in real time. In the context of education, researchers have identified a range of AI-enabled cognitive affordances that may shape student learning (Järvelä, Zhao, Nguyen, & Chen, 2025; Labadze, Grigolia, &

Machaidze, 2023; Luckin, Holmes, Griffiths, & Forcier, 2016; Mavrikis & Holmes, 2019). These include:

(1) Personalization: tailoring explanations, examples, or tone

(2) Interactivity: responding to prompts and refining outputs

(3) Convenience: on-demand support, summarization, formatting

(4) Adaptive feedback: revision suggestions or tone adjustments

(5) Prediction and profiling: anticipating needs and suggesting next steps

(6) Social presence: engaging in simulated dialog

(7) Competition: using AI to improve performance relative to peers

AI's value depends on cognitive preferences: Where some need structure, others clarity. Recognizing this is the key to understanding how AI shapes learning behaviors. Figure 1 illustrates the relationship between student cognitive learning preferences and common affordances offered by generative AI tools. Cells highlighted in green indicate affordances likely to be perceived as supportive by students with the corresponding preference. Red-outlined cells indicate potential areas of risk, where over-reliance on convenience-oriented or competitive AI features may undermine critical thinking or independent learning processes. In Figure 1, "risk" does not imply that an affordance is inherently harmful; it signals a predictable failure mode when the affordance systematically reduces necessary cognitive work. For example, for structural thinkers, high-convenience structuring can externalize planning to the tool; for verbal learners, fluent explanations can substitute for generative processing; and for performance-oriented learners, polishing/benchmarking features can reward surface completion over mastery. The risk markers therefore indicate where design should add counterweights (e.g. justification prompts, process checks, or AI-free steps) rather than where use should be prohibited.

It is worth noting that *Convenience* is a column that is highlighted for both its positive and negative influence across all cognitive preferences. To elaborate further, each cognitive

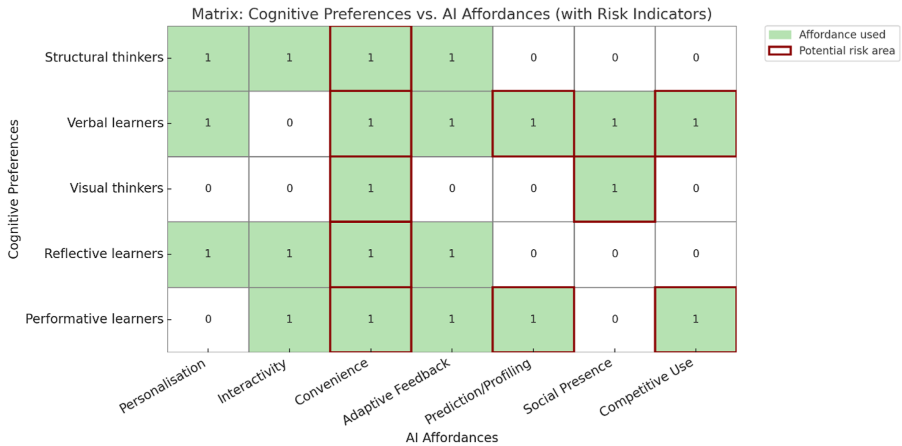

Figure 1. Matrix of cognitive preferences vs. AI affordances (with risk indicators). Note. Risk markers are included based on existing concerns in the literature about reduced student effort, overdependence, or surface-level engagement (Aggarwal, 2023; Bhavani *et al.*, 2025; Chen, 2020; Domshlak, Hüllermeier, Kaci, & Prade, 2011; Holmes, 2020; Huang, Peng, & Teo, 2025; Järvelä *et al.*, 2025; Jian, 2023; Shiohira & Holmes, 2023).
Source: Authors' own work

preference category can be examined individually. For structural thinkers, AI helps by organizing content into steps or outlines, which supports their need for order and clarity (Kanchon, Sadman, Nabila, Tarannum, & Khan, 2024; Susilo, 2024). However, this can also reduce their ability to plan independently when structure is not provided (Ashton & Franklin, 2022; Shiohira & Holmes, 2023). Verbal learners benefit from tools that rephrase text or explain ideas through language, which fits their preference for reading and writing (Kim & Kim, 2022; Mayer & Massa, 2003). Conversely, they may rely too much on AI explanations and avoid deeper thinking (Bhavani, Shetty, & DM, 2025; Wang *et al.*, 2023). Visual learners often use AI to create diagrams or tables that help them understand relationships (Kim & Kim, 2020; Xue *et al.*, 2023), but they may stop practicing how to create these visuals themselves, which is important for deeper understanding (Bower, 2008; Mayer & Massa, 2003). Reflective learners benefit from the ability to get AI feedback at their own pace, which supports thoughtful learning (Essa, Çelik, & Human-Hendricks, 2023; Järvelä *et al.*, 2025). However, the instant answers from AI may prevent them from sitting with uncertainty, which is important for critical thinking (Ashton & Franklin, 2022; Kopecka, Such, & Luck, 2024). Learners with a strong performance orientation (here defined as prioritizing speed, grades, and appearing competent rather than mastery and strategy development) may find AI helpful for producing polished results (Baltezarević & Baltezarević, 2024; Castro, Chiappe, Rodríguez, & Sepulveda, 2024), but the same affordances can also increase shallow engagement and shortcut use (Cameron & Rideout, 2020; Wang *et al.*, 2023). To further unpack this relationship, the following section explores the underlying mechanisms of amplification in more depth.

AI as a learning preference amplifier

Mechanisms of amplification

The role of AI in personalized learning has taken strides forward as many systems now are designed to detect, respond to, and adapt based on individual learning preferences (Baltezarević & Baltezarević, 2024; Jauhiainen & Garagorry Guerra, 2024; Susilo, 2024). This amplification process where AI tools reinforce or support students' preferred cognitive strategies can occur through technical and pedagogical mechanisms (Castro *et al.*, 2024; Lokare & Jadhav, 2023). Figure 2 illustrates the technical and pedagogical mechanisms through which AI systems detect, respond to, and amplify student learning preferences. On the left, detection and categorization processes include behavioral data collection, pattern recognition via machine learning, and

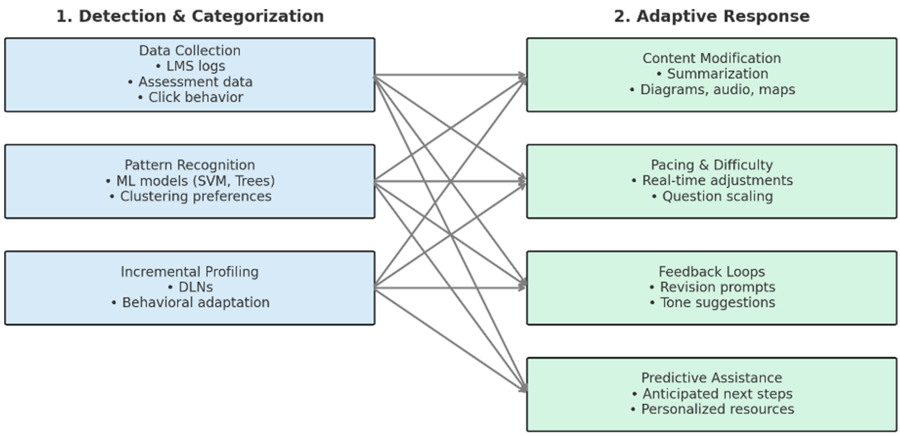

Figure 2. Mechanisms of LLM-based learning-preference amplification. Source: Authors' own work

incremental learner modeling (Essa *et al.*, 2023; Bhavani *et al.*, 2025). Adaptive responses personalize content, pacing, and feedback to anticipate student needs (Chen, 2020; Jian, 2023; Lee & Moore, 2024). These mechanisms align with cognitive tendencies, shaping engagement and development (Kim & Kim, 2020; Mayer & Massa, 2003). They are not neutral; they scaffold structure-seekers, aid visual thinkers, and prompt reflection (Conole & Dyke, 2004; Halkiopoulos & Gkintoni, 2024). These mechanisms form a self-reinforcing loop that can support, constrain, or cement learning habits over time, depending on how students recognize and use the possibilities offered by AI, and how educators design learning around those possibilities (Faraj & Azad, 2012; Xue *et al.*, 2023).

Detecting and categorizing learning preferences

AI-driven personalization starts by analyzing behavioral data, such as quiz results and engagement timing, to classify students into cognitive preference types (Essa *et al.*, 2023; Ezzaim, Dahbi, Aqqal, & Haidine, 2024). More advanced systems use developmental learning networks (DLNs) and deep learning techniques to continuously refine learner profiles, updating their understanding based on ongoing student interactions (Kim & Kim, 2022). This reflects what AT calls "affordance activation": students notice and engage with the features of AI that align with their own cognitive preferences. As they repeatedly use the tool in these ways, the system becomes more responsive to those patterns, thereby creating a feedback loop in which the AI not only adapts to the student, but the student increasingly adapts to the AI's most accessible affordances.

This loop offers significant potential benefits. It can increase efficiency, reduce cognitive load, and foster a sense of mastery, especially for learners who previously struggled to find their footing in rigid or standardized environments. By shaping learning to feel more intuitive, AI may help students stay engaged longer and move through material more confidently. However, the same loop may also *narrow* the scope of learning. As students gravitate toward familiar affordances (i.e. summarization or simplification), they may avoid more cognitively demanding tasks like synthesis, critical comparison, or ambiguity tolerance. Over time, this can lead to a kind of preference ossification, where the student's learning behavior could become rigid, shaped more by what the AI makes easy than by what is developmentally valuable. In this way, the affordance loop can shift from being a scaffold to becoming a ceiling. Self-regulated learning (SRL) makes this "scaffold-versus-ceiling" dynamic more precise. In SRL terms, amplification is beneficial when AI support strengthens learners' monitoring and control (e.g. planning, checking understanding, adjusting strategies), but harmful when it replaces those processes and reduces calibration. This aligns with SRL models in which outcomes depend not only on cognitive tendencies, but on learners' ability to select, evaluate, and revise strategies over time (Pintrich, 2000; Winne & Hadwin, 1998; Zimmerman, 2002). Accordingly, the same affordance (e.g. summarization) can either support SRL (when used as a check followed by revision) or undermine SRL (when used as an endpoint that closes inquiry).

Figure 3 operationalises how generative AI can amplify students' cognitive preferences by aligning system affordances with individual learning tendencies. Drawing on AT, the model maps three core domains: cognitive preferences, AI affordances, and learning outcomes. The overlapping areas signal critical sites of interaction. *Affordance Activation* marks the intersection where student traits (e.g. structural or visual thinking) and AI features (e.g. diagrams, structured outputs) align, enabling tools to resonate more deeply with learners' inherent styles (Essa *et al.*, 2023; Mayer & Massa, 2003). *Design Potential* denotes the intersection of AI capabilities and educational objectives, identifying where scaffolded prompting, feedback design, and task calibration support learning in adaptive and pedagogically meaningful ways (Kanchon *et al.*, 2024; Bhavani *et al.*, 2025). By contrast, *Overuse Risks* captures the space where student preferences and learning outcomes drift into less productive engagement, potentially reinforcing inflexible strategies or surface-level engagement (Ashton & Franklin, 2022; Kirschner & Van Merriënboer, 2013). At the center lies the *Amplification Zone*, where all three domains meet. This is the point of highest leverage

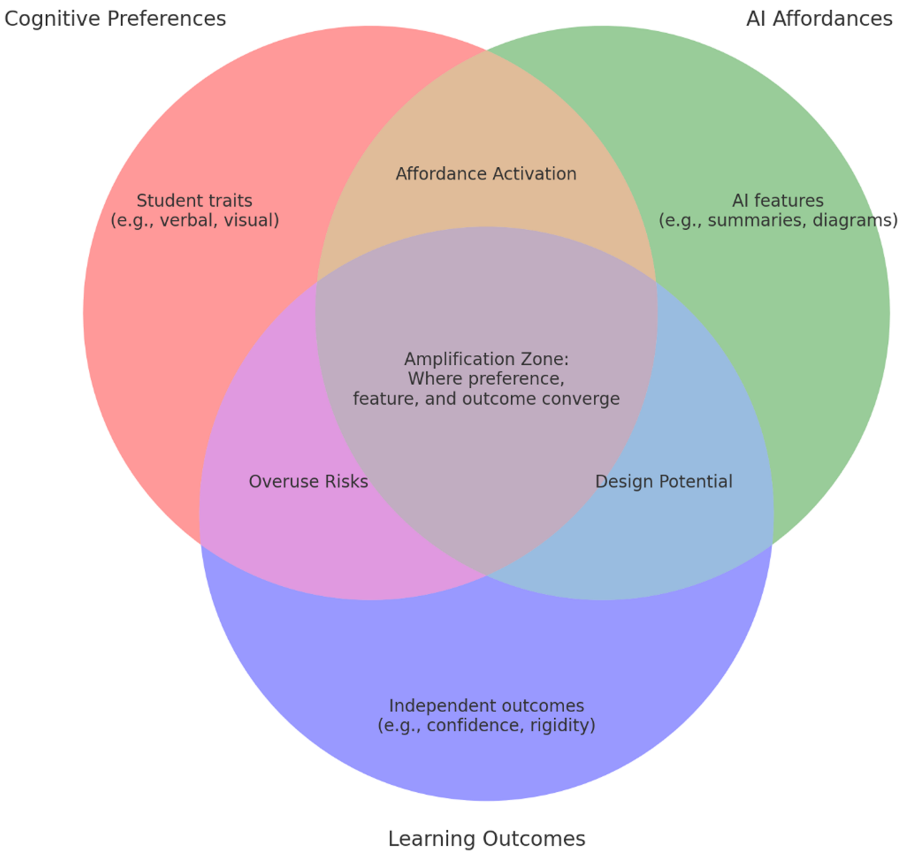

Figure 3. Amplification zone: Venn model of cognitive preferences, LLM affordances, and learning outcomes. Note. This model is based on a synthesis of literature on AI affordances, cognitive preferences, and personalized learning. Source: Authors' own work

and highest risk—capable of enabling transformation or entrenchment depending on pedagogical choices (Aggarwal, 2023; Almusaed, Almssad, Yitmen, & Homod, 2023). The figure therefore translates the theoretical argument into an actionable lens for design, clarifying the interplay between personalization, learner characteristics, and the development of learning outcomes. Educators now face a double-edged sword. They must balance the empowering potential of AI with the need for caution to prevent student over-reliance - mirroring Daedalus's timeless dilemma: how to enable flight without courting collapse.

Deriving testable propositions

Building on the AI Amplification Zone model and the mechanisms outlined above (preference detection → adaptation → ossification), the following testable propositions specify when and how LLM–preference alignments are likely to promote deeper learning and when they risk maladaptive reinforcement.

(P1) When tasks embed desirable difficulties, LLM support shift learners toward deeper strategies despite initial preference mismatch.

(P2) Scaffolded prompting plus human–AI co-teaching reduces over-reliance among learners with a strong performance orientation (i.e. prioritizing speed, grades, and polished outputs over mastery)

(P3) Periodic AI-free checkpoints disrupt ossification loops and preserve transfer.

Together, these propositions delineate a research agenda: they can be operationalized as falsifiable hypotheses (e.g. randomized task designs with/without desirable difficulties; human–AI co-teaching vs. AI-only feedback; scheduled AI-free checkpoints) and evaluated with behavioral, performance, and transfer measures. Here, performance orientation' is *not* treated as a stable learner type; it is a situationally variable goal orientation that can co-exist with any cognitive preference profile and can be intensified by assessment signals, time pressure, and tool affordances that reward polish and speed (Pintrich, 2000).

Operational boundaries

For empirical testing, we treat (1) *desirable difficulties* as task features that require generative processing (e.g. retrieval, comparison, justification, or modality switching) rather than mere fluency; (2) *scaffolded prompting* as structured prompts that require learners to articulate goals, assumptions, reasoning steps, and uncertainty; (3) *human–AI co-teaching* as an explicit division of feedback roles (e.g. AI for formative drafting/structure; instructor for conceptual validity, standards, and transfer); and (4) *AI-free checkpoints* as scheduled segments where key reasoning steps must be completed without AI support and assessed for explanation and transfer. Outcomes should include not only performance, but SRL-relevant process indicators (planning quality, monitoring accuracy, revision behavior) and transfer (near/far) (Winne & Hadwin, 1998; Zimmerman, 2002).

Personalization and engagement

Once a learner's preferences are detected or inferred, AI systems begin to personalize the learning experience. They do so by adjusting not only what content is delivered, but how, when, and in what tone. These personalized responses can feel affirming, fostering recognition and support with the potential to increase perceived self-efficacy—particularly during the uncertainty of first-year transition. For first-year students navigating the uncertainty of university transition, perceived alignment with AI can enhance emotional and cognitive engagement (Castro *et al.*, 2024; Halkiopoulos & Gkintoni, 2024). Personalization reduces anxiety, increases motivation, and supports autonomy when learning feels tailored to individual needs (Castro *et al.*, 2024; Halkiopoulos & Gkintoni, 2024; Kanchon *et al.*, 2024). These benefits often justify AI adoption in education (Hooshyar *et al.*, 2024). However, personalization also fosters cognitive fluency (learning feels easier when matched to preferred modalities), which can reduce cognitive load but risk overfitting. A key design lever is metacognitive scaffolding: prompts that require learners to articulate goals, justify steps, and evaluate understanding rather than simply accept fluent outputs. Such scaffolds help ensure that preference–affordance "fit" supports monitoring and strategy control, reducing the risk that fluency becomes a substitute for learning. Verbal learners may rely on summaries, avoiding complex visuals; visual learners may prefer diagrams over analytical writing. This can create a "greenhouse effect": optimal for early growth but limiting later flexibility. Educators must balance personalization with increasing desirable difficulty to support long-term learning. Moreover, personalization may affect different dimensions of engagement differently:

(1) Cognitive engagement: increased focus and organization

(2) Emotional engagement: reduced anxiety, increased motivation

(3) Behavioral engagement: more frequent and sustained participation

(4) Social engagement: potentially decreased if AI replaces human interaction

The tension between support and avoidance reinforces the model's central argument: AI does not level the playing field – it reshapes it, and often reinforces existing patterns of cognition and behavior. Figure 3 highlights the Amplification Zone as the ideal point of convergence where cognitive preference, AI affordance, and learning outcome align. But this convergence doesn't equalize opportunities, rather it amplifies what students already tend toward.

Despite their promise, key challenges remain. First, more empirical research is needed to compare deep learning models for preference detection (Ezzaim *et al.*, 2024). Second, ethical concerns persist around profiling, transparency, and bias, particularly in formative assessment (Ashton & Franklin, 2022; Kopecka *et al.*, 2024; Pan, Biegley, Taylor, & Zheng, 2024). Finally, AI must not replace productive intellectual struggle essential to deep learning.

Discussion

The discussion focuses on where the Amplification Zone can be leveraged to promote productive stretch and where guardrails are required to prevent strategy ossification across first-year learning contexts. It synthesizes implications for first-year pedagogy, assessment design, and future empirical research.

System-level implications

Because amplification operates through repeated learner–tool interactions, it is not only a classroom issue but also a programme and institutional system issue. Course teams may need shared norms for AI use (what is permitted, when, and why), staff development to support human–AI feedback role clarity, and assessment policies that reward process evidence (e.g. drafts, rationales, and checkpoint justifications) rather than polished outputs alone. At scale, institutions also face governance questions about profiling and transparency (what data are inferred, how students can opt out, and how bias is monitored), making the AI Amplification Zone relevant to curriculum policy, academic integrity design, and learning analytics practice, not only to individual teaching tactics.

Implications for first-year education

The following scenarios are illustrative vignettes used to make the proposed mechanisms concrete; they do not report new empirical findings and should be read as design-relevant examples that motivate the testable propositions (P1–P3).

Because the AI Amplification Zone is a new conceptual model, the implications are presented as testable design hypotheses mapped to stages of the amplification mechanism (detection → adaptation → ossification), rather than as evidence-backed best practices.

The transition from secondary to higher education places significant cognitive and emotional demands on first-year students, who must quickly adapt to new environments and academic expectations (Cameron & Rideout, 2020). These challenges can heighten anxiety, especially when traditional instruction fails to support students' preferred learning styles (Babyuk *et al.*, 2019; Lujan & DiCarlo, 2006). AI can scaffold the first-year adjustment period by amplifying familiar cognitive strategies—sometimes productively, sometimes at the cost of flexibility.

This illustrative vignette about qualitative coding provides an example. In a first-year qualitative research task, learners may use generative AI to support open coding of interview excerpts, particularly when the bottleneck is articulating emerging patterns in appropriate academic language. Used as a *drafting partner* (for example, by generating candidate codes that students then justify against the raw data), the tool can make the interpretive phase feel more accessible and reduce anxiety during early transition (Aggarwal, 2023; Kanchon *et al.*, 2024). Used as a *substitute analyst*, however, it can also encourage premature closure if AI-generated codes are adopted without evidential checking or revision. Another iIllustrative vignette concerns design prototyping. In a design thinking assignment, AI-generated wireframes and draft personas can lower technical barriers and increase students' willingness to attempt more ambitious prototypes, thereby expanding creative confidence. At the same time, profiling, prediction, and competitive affordances may disproportionately reward speed and polish, reinforcing a performance orientation if assessment signals privilege outputs over reasoning and process (Ashton & Franklin, 2022; Babyuk *et al.*, 2019; Bhavani *et al.*, 2025; Castro *et al.*, 2024). Together, these vignettes illustrate how the same affordances can expand capability or mask shallow processing depending on task design and assessment constraints.

However, as the proposed model and matrix illustrate, over-reliance on these features can promote surface learning, conformity, and cognitive avoidance—limiting flexibility and adaptive growth (Halkiopoulos & Gkintoni, 2024; Kopecka *et al.*, 2024; Wang *et al.*, 2023). To counteract this, educators should pair AI-supported personalization with "productive struggle" tasks that stretch students beyond their comfort zones (Shiohira & Holmes, 2023). In this way, AI becomes not just a shortcut, but a designable tool: useful when paired with constraints that preserve explanation, transfer, and productive struggle. Table 2 illustrates how the conceptual model (Figure 3) can inform concrete interventions that address both the potential and pitfalls of AI in early-stage higher education.

Drawing on the authors' experience in higher education teaching, a plausible pattern we have recurring observed informally is that AI can accelerate production of polished artefacts while weakening students' visibility into *process*. For example, in a design thinking assignment, student teams may generate personas, journey maps, and prototypes with AI while doing minimal user research; outputs appear complete, yet teams struggle to justify decisions with evidence or to articulate how user data informed design choices. In response, a "reclaim-

Table 2. Venn zone interventions and design actions

Venn zone	Opportunity/ intervention	Potential risk	Possible design action
Cognitive preferences only	Offer self-assessment tools to help students identify their cognitive preferences early in the semester	Students may remain unaware of helpful AI features, limiting tool effectiveness	Incorporate a short, reflective learning preferences quiz linked to suggested AI tools
AI affordances only	Provide guided tours or tutorials to help students discover various AI features and affordances	AI use may be generic or ineffective if not aligned to student needs	Embed AI tooltips and contextual help within LMS platforms to highlight affordances
Learning outcomes only	Design formative assessments that track shifts in cognitive behavior and awareness over time	Educators may misattribute success to AI use without tracking cognitive engagement	Create diagnostic tasks that prompt students to explain how they approached a problem
Affordance activation (preferences + AI)	Use personalized AI support to build initial confidence and scaffold skill development	Students may become over-reliant on specific supports, reducing cognitive flexibility	Include periodic oral defenses or whiteboard checks where appropriate to probe understanding
Overuse risks (preferences + outcomes)	Introduce reflection prompts or scaffolds to interrupt overly habitual AI use	Narrow personalization may inhibit resilience, adaptability, or transfer of skills	Schedule structured AI-free moments in class or variation challenges to promote adaptability

Note(s): Opportunities and Risks of the AI Amplification Model, as well as possible design actions
Source(s): Authors' own work

the-process" step can be introduced: teams must critique AI-generated artefacts against primary data (e.g. interview excerpts) and rebuild key decisions with explicit justification, repositioning AI as a starting point rather than a substitute for user-centered reasoning. A similar dynamic can occur in quantitative tasks, where groups may accept AI-suggested statistical tests without understanding assumptions, producing reports that look sophisticated but contain methodological errors. Here, an AI-free checkpoint (e.g. a brief oral defense of test choice and assumptions, or a short hand-worked verification step) shifts assessment from presentation to reasoning. Taken together, this vignette illustrates the paper's core design claim: AI can amplify either productive strategy development or shortcut behavior depending on whether tasks and assessments preserve explanation, evidential grounding, and metacognitive control.

Empirical validation of amplification patterns

While the idea that AI can amplify students' cognitive preferences is conceptually strong, there remains a lack of longitudinal and experimental evidence assessing its long-term effects on learning. Future research should explore whether this amplification leads to deeper learning or whether it risks reinforcing rigid-thinking patterns over time.

Validating the model through mixed-methods and cross-institutional studies would allow educators to differentiate between healthy personalization and cognitive overfitting. This would then help tailor AI strategies to learner development stages with greater nuance and accountability (Bhavani *et al.*, 2025; Essa *et al.*, 2023; Ezzaim *et al.*, 2024; Wang *et al.*, 2023).

Designing for flexibility, not just fit

Personalization enhances short-term engagement, but risks narrowing learners' capacity to adapt. Educational design must include features that challenge students to work outside their preferred modalities, which Bjorn and Bjork (2011) describe as "desirable difficulties." By intentionally embedding moments of cognitive stretch (e.g. switching modalities, requiring reflection on unfamiliar representations), AI systems can help students not only feel competent but also become more cognitively agile over time (Bower, 2008). This opens the door to personalization as a growth mechanism, rather than just a comfort feature.

Ethical profiling and transparency

AI-driven personalization relies on profiling, which introduces critical concerns around data ethics, bias, and opacity. Learners must remain informed participants in this process rather than passive subjects of hidden algorithms. With ethical design, AI can become a tool for empowering student agencies. Transparent dashboards, adjustable profiles, and opt-in personalization allow students to co-author their learning pathways, which could in turn, foster metacognitive awareness and trust in their educational system (Ashton & Franklin, 2022; Kopecka *et al.*, 2024).

Human-AI Co-teaching models

AI should not replace educators but extend their reach. Human-AI collaboration in teaching offers the opportunity to combine scalable, data-driven support with relational insight and pedagogical interpretation. In co-teaching models, AI can manage routine personalization and early detection of disengagement, freeing educators to focus on high-impact relational work, such as mentoring, feedback, and fostering critical thinking (Almusaed *et al.*, 2023; Järvelä *et al.*, 2025; Létourneau *et al.*, 2025).

Ensuring equity and inclusive design

Without care, personalization tools may inadvertently deepen educational inequalities, favoring learners whose styles or contexts align with dominant design assumptions. Developing students'

AI literacy is therefore a parallel priority for equitable, transparent use (Almatrafi, Johri, & Lee, 2024). Inclusive design that is guided by the principles of Universal Design for Learning (UDL) could ensure that AI supports the broadest range of learners. Adaptive interfaces, multimodal options, and culturally sensitive content scaffolding have the potential to transform AI into a powerful equalizer rather than a divider (Babyuk *et al.*, 2019; Wang *et al.*, 2023).

Conclusion

As a conceptual/theory-building synthesis rather than an empirical study, this work integrates Affordance Theory with cognitive–motivational research to specify mechanism-level links and a testable framework for first-year contexts. The model clarifies when preference–affordance alignments enable productive stretch versus ossify strategy use, offers a tractable lens for instructional design, and invites empirical evaluation of the propositions and refinement of the Amplification Zone across diverse first-year settings. This paper argues that AI does not merely *respond* to student preferences, but actively *shapes* them. Through repeated interactions, students come to rely on the affordances that best match their cognitive comfort zones, while the AI becomes increasingly responsive to those same patterns. This feedback loop, what this paper has termed *amplification*, can enhance fluency and engagement, but may also narrow cognitive flexibility over time.

The proposed *AI Amplification Model* maps the interaction between cognitive preferences, AI affordances, and learning outcomes. In doing so, it clarifies how students' behavioral tendencies and learning environments co-construct each other. This paper offers a visual and theoretical tool to support educators in recognizing the zones of benefit, risk, and design potential. When applied to first-year bachelor education, this model underscores a pressing need: to design with awareness of cognitive stretch and lean into desirable difficulty.

The transition into university often brings a sharp increase in cognitive load and emotional pressure. In this context, personalization offers immediate relief, but at a potential long-term cost. For educators, the task is not to eliminate AI personalization, but to wield it strategically: scaffolding new skills, expanding learners' cognitive repertoires, and nudging them toward less-familiar but developmentally necessary forms of engagement. Ultimately, AI may reshape the playing field, but educators remain the architects. Accordingly, the Amplification Zone provides a tractable lens for research and design, distinguishing supportive personalization from cognitive overfitting and specifying where interventions should target flexibility rather than comfort. If Daedalus's dilemma teaches us anything, it is this: wings alone are not the danger. It is in how they are used and where we aim to fly.

Future research

As a conceptual synthesis, the claims made await empirical testing, for which initial designs are outlined in *Future research* (P1–P3). The propositions advanced above specify a program for empirical testing. For example, P1 (desirable difficulties) can be examined via randomized or crossover designs that compare LLM-supported vs. unsupported practice under calibrated difficulty; P2 (co-teaching) via factorial designs manipulating human/LLM feedback roles; and P3 (AI-free checkpoints) via interrupted-use protocols to assess near/far transfer. Outcomes should include process (prompt traces, time-on-task, metacognitive reports), performance (accuracy, retention), and equity indicators (heterogeneous treatment effects across student subgroups).

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Further reading

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